Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [10]
- Focus on image analysis component [3]

What is the purpose of this section?

- What is plant phenotyping?
- Discuss what can be measured by plant phenotyping?
- Why bother about plant phenotyping?

Context for plant phenotyping? It is of vital importance in plant breeding to be able to propogate desirable heritable traits from plants through controlled human action [17]. This is especially important in the context of food security where a cultivars resistant to adverse abiotic and biotic stresses affect the final overall crop yield. The literature indicates that plant phenotyping has become an effective tool in plant breeding methods. [1].

What is plant phneotyping Plant phenotyping is then the evaulation of phenotypes in plants with the main goal of producing quantative data to study the dynamic interaction between a plant and its environment [4].

Definitons 2
The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [6]

Definitons 3
For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture, and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [4]

Definitons 4
The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [16]

Definitons 5
Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [13]

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [8]. Who stated that a phenotype is all types of organism which are idenfitied from one antoher by inspection or by some measuremment. So it is an obserable primitive trait or a complex trait of a plant that can be identified visually or measured and evaulated numerically. Therfore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various diffrent growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant plantphenotyping suffers from scalabality issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogensis of a cultivar. However in the past decade image processing techniques have been applied successfully to aquire phenotypical data in a non-destructive matter [2].. There exists many different image techniques that have been applied in greenhouses and in the field which allow for the assesment of resistance to disease, biomass, soli salinity, etc using various differenr sensors, and techniques.

What affects plant growth?
Plant biomass is affected by stresses and disturbances [5]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests.

How can we measure plant stresses and disturbances?
Stresses and disturbances of plants are induced by environmental conditions the organisms are placed in, and therefore transitively induce phenotypical traits. We can exploit this fact by measuring plant phenotypical traits, we can infer the prescence of potential stresses and disturbances on cultivars. For example thermal imaging allows to meausre the phenotype of plant temperature or plant transpiration, which in turn can give us an indication of soil salinity. We indirectly infer environmental conditions based on phenotypical traits

What physical traits of a plant can be measured phenotypically?

Conceptual basis Typically in the context of plant breeding we care about plant yield, resistant to abiotic and biotic stresses. That is in general we seek plants which are fit. In the literature of the term trait is used to describe individual scoped properties, communit scoped properties end even environmental properties. trait is any morphological, physiological or phenological feature measurable at the individual level, from the cell to the whole-organism level, without reference to the environment [26]. Plant phenotyping reflects this in its achievements and developments over the past decade. The key is environmental related performance metrics are derived from individual plant traits. So these phenotypical traits act as proxies that indirectly preditc plant fitness and performance.

Morphological traits About plant sturcture, which affects its performance and gives indication regarding its health

- Leaf width
- Number of leaves
- Leaf inclination angle
- Leaf area
- Root system architecture (e.g root diameter)
- Plant shoots
- Fruit
- Flowers

Physiological traits About how plant operates in repsonse to its growing conditions

- Stomotal conductance
- Chlorophyll content of leaf
- Plant water status
- Plant photosynthesis (metablism)
- Nutrient content

Phenological traits These are phenological traits (when certain biological events occur that depend on climates)

- When fruit or flowers are yielded
- Yield prediction

The use of plant phenotyping and detecting plant stress Plant biomass is affected by stresses and disturbances [5]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests

Biotic Stress + Disturbance.

- Pathogenic organisms
- Pests
- Parasitc plants

Abiotic Stress + Disturbance

- Drought
- Flood
- Salinity
- Heat Stress
- Raditation Stress, light stress

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquistion techniques
- Discuss image processing techniques used in plant phenotyping

How does image processing relate to plant phenotyping?
Fill out this section

The conceptual pipeline for image processing techniques in plant phenotyping
This paragraph should discuss how image processing is applied. I.e it should map the process from image acquistion to image processing to data generation/analysis

Foundations of image acquistion in plant phenotyping

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [13]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [13]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [13].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [13]

Image acquistion methodologies in plant phenotyping

Thermal imaging sensors

- Conceptual basis of this technique [19]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.
- By inferring transpiration rate of leaves which is governed by stomoatal conductance.
- Stomotal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [7]

Hyperspectral sensors

- Based on the principles of spectroscopy. Can detric biochemical, morphological and physilogical traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about clholrphyll content. [22].
- Conceptual basis and applications [21]
- Crop yield
- Biotic and abiotic factors

Tomography

- three-dimensional visualization of plant internal structures, root systems, and physiological processes
- Based on principles of xray cts, moving sensor takes planar cross sections to 3d visialize plant
- Drought stress [25]
- Yield Estimations
- has references for usages [18]

Flouresnce sensors

- Conceptual basis of this technique [23]
- Predict metabolism using photosynthesis which tells us about growth status
- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorphyll content canbe us used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [14]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Acitinic light involved here?

RGB Camera

- Mimic human vision
- measure plant shoot biomass (yield potential) [13]
- leaf senance (tissue resistance to salt)

What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

What is the conceptual basis of low-cost automated greenhouses

- Germplasm Evaluation
- phenotyping platforms
- phenyping image acquistion
- phenotyping image preprocessing

Current applications

- Often combined with machine learning algorithms
- Cheap Rasberry PI with RGB camera [24]
- Autonomous robots with sensors and cameras [20]
- use off the shelf commerical software like microsoft xbox360 kinect [12]

What affects a low-cost automated greenhouse cost
Points from this paper [9]

- Image acquistion technique (imaging system)
- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- enery cost of technology
- amount of data generated, where is it stored, how frequently? [3]

Challenges of low-cost automated greenhouses

- Many phenotyping systems exisiting are comercially developed under patent so hardware and software can not be modified. [11]
- Current applications use moving conveyor belt (benchtop) [11]
- or convery belt [11]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [20]
- large amounts of data need to be colleceted, and processed. Exisitng phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for labratory systems, but not extended for real applications. Standarides off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [15]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [27] suggesting need for low-cost systems that can support a wider variety of genetypes and phenotypical traits.

What does a typical preprocessing pipeline look like?

- automated image classification (using machine learning)

Fill out the conclusion section.

References

[1] Achala Anand, Madhumitha Subramanian, and Debasish Kar. "Breeding Techniques to Dispense Higher Genetic Gains". In: *Frontiers in Plant Science* 13 (Jan. 19, 2023), pp. 4–5. ISSN: 1664-462X. DOI: 10.3389/fpls.2022.1076094. URL: https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.1076094/full (visited on 05/16/2025).

[2] Jayme Garcia Arnal Barbedo. "Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases". In: *SpringerPlus* 2.1 (Dec. 7, 2013), p. 660. ISSN: 2193-1801. DOI: 10.1186/2193-1801-2-660. URL: https://doi.org/10.1186/2193-1801-2-660 (visited on 05/11/2025).

[3] Jianjun Du et al. "Greenhouse-Based Vegetable High-Throughput Phenotyping Platform and Trait Evaluation for Large-Scale Lettuces". In: *Computers and Electronics in Agriculture* 186 (July 2021), p. 106193. ISSN: 01681699. DOI: 10.1016/j.compag.2021.106193. URL: https://linkinghub.elsevier.com/retrieve/pii/S0168169921002106 (visited on 05/24/2025).

[4] Fabio Fiorani and Ulrich Schurr. "Future Scenarios for Plant Phenotyping". In: *Annual Review of Plant Biology* 64.1 (Apr. 29, 2013), pp. 267–291. ISSN: 1543-5008, 1545-2123. DOI: 10.1146/annurev-arplant-050312-120137. URL: https://www.annualreviews.org/doi/10.1146/annurev-arplant-050312-120137 (visited on 05/14/2025).

[5] J.P Grime. "Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory". In: *The American Naturalist* 111.982 (1997), p. 1169. DOI: 10.1086/283244. URL: https://www.journals.uchicago.edu/doi/abs/10.1086/283244 (visited on 05/19/2025).

[6] Qiaosheng Guo and Zaibiao Zhu. "Phenotyping of Plants". In: *Encyclopedia of Analytical Chemistry*. John Wiley & Sons, Ltd, 2014, pp. 1–15. ISBN: 978-0-470-02731-8. DOI: 10.1002/9780470027318.a9934. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470027318.a9934 (visited on 05/14/2025).

[7] Roselyne Ishimwe, K. Abutaleb, and F. Ahmed. "Applications of Thermal Imaging in Agriculture—A Review". In: *Advances in Remote Sensing* 03.03 (2014), pp. 128–140. ISSN: 2169-267X, 2169-2688. DOI: 10.4236/ars.2014.33011. URL: http://www.scirp.org/journal/doi.aspx?DOI=10.4236/ars.2014.33011 (visited on 05/19/2025).

[8] W. Johannsen. "The Genotype Conception of Heredity". In: *The American Naturalist* 45.531 (Mar. 1911), pp. 129–159. ISSN: 0003-0147, 1537-5323. DOI: 10.1086/279202. URL: https://www.journals.uchicago.edu/doi/10.1086/279202 (visited on 05/14/2025).

[9] Yonghyun Kim, Jinyoung Y. Barnaby, and Scott E. Warnke. “Development of a Low-Cost Automated Greenhouse Imaging System with Machine Learning-Based Processing for Evaluating Genetic Performance of Drought Tolerance in a Bentgrass Hybrid Population”. In: *Computers and Electronics in Agriculture* 224 (Sept. 1, 2024), p. 108896. ISSN: 0168-1699. DOI: 10.1016/j.compag.2024.108896. URL: https://www.sciencedirect.com/science/article/pii/S0168169924002874 (visited on 05/24/2025).

[10] R. H. Laxman et al. “Non-Invasive Quantification of Tomato (Solanum Lycopersicum L.) Plant Biomass through Digital Imaging Using Phenomics Platform”. In: *Indian Journal of Plant Physiology* 23.2 (June 1, 2018), pp. 369–375. ISSN: 0974-0252. DOI: 10.1007/s40502-018-0374-8. URL: https://doi.org/10.1007/s40502-018-0374-8 (visited on 05/24/2025).

[11] Daoliang Li et al. “High-Throughput Plant Phenotyping Platform (HT3P) as a Novel Tool for Estimating Agronomic Traits From the Lab to the Field”. In: *Frontiers in Bioengineering and Biotechnology* 8 (Jan. 13, 2021), p. 623705. ISSN: 2296-4185. DOI: 10.3389/fbioe.2020.623705. URL: https://www.frontiersin.org/articles/10.3389/fbioe.2020.623705/full (visited on 05/24/2025).

[12] Dawei Li et al. “Digitization and Visualization of Greenhouse Tomato Plants in Indoor Environments”. In: *Sensors* 15.2 (2 Feb. 2015), pp. 4019–4051. ISSN: 1424-8220. DOI: 10.3390/s150204019. URL: https://www.mdpi.com/1424-8220/15/2/4019 (visited on 05/24/2025).

[13] Lei Li, Qin Zhang, and Danfeng Huang. “A Review of Imaging Techniques for Plant Phenotyping”. In: *Sensors (Basel, Switzerland)* 14.11 (Oct. 24, 2014), pp. 20078–20111. ISSN: 1424-8220. DOI: 10.3390/s141120078. PMID: 25347588. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4279472/ (visited on 04/20/2025).

[14] Hartmut K. Lichtenthaler and Ursula and Rinderle. “The Role of Chlorophyll Fluorescence in The Detection of Stress Conditions in Plants”. In: *C R C Critical Reviews in Analytical Chemistry* 19 (sup1 Jan. 1, 1988), S29–S85. ISSN: 0007-8980. DOI: 10.1080/15476510.1988.10401466. URL: https://doi.org/10.1080/15476510.1988.10401466 (visited on 05/22/2025).

[15] Massimo Minervini, Hanno Scharr, and Sotirios A. Tsaftaris. “Image Analysis: The New Bottleneck in Plant Phenotyping”. In: *IEEE Signal Processing Magazine* 32.4 (July 2015), pp. 126–131. ISSN: 1053-5888, 1558-0792. DOI: 10.1109/MSP.2015.2405111. URL: https://ieeexplore.ieee.org/document/7123050/ (visited on 05/24/2025).

[16] Semira Ebrahim Mohammed. “Review on Plant Phenotyping”. In: (). URL: https://www.abrinternationaljournal.org/abstract/review-on-plant-phenotyping-1102322.html (visited on 05/05/2025).

[17] Thomas J. Orton. "Introduction". In: *Horticultural Plant Breeding*. Elsevier, 2020, pp. 3–7. ISBN: 978-0-12-815396-3. DOI: 10.1016/B978-0-12-815396-3.09986-8. URL: https://linkinghub.elsevier.com/retrieve/pii/B9780128153963099868 (visited on 05/16/2025).

[18] Stefan Paulus. "Measuring Crops in 3D: Using Geometry for Plant Phenotyping". In: *Plant Methods* 15.1 (Sept. 3, 2019), p. 103. ISSN: 1746-4811. DOI: 10.1186/s13007-019-0490-0. URL: https://doi.org/10.1186/s13007-019-0490-0 (visited on 05/23/2025).

[19] Anupma Prakash. "THERMAL REMOTE SENSING: CONCEPTS, ISSUES AND APPLICATIONS". In: *International Archives of Photogrammetry and Remote Sensing* 33 (2000), pp. 239–243.

[20] Amine Saddik et al. "Mapping Agricultural Soil in Greenhouse Using an Autonomous Low-Cost Robot and Precise Monitoring". In: *Sustainability* 14.23 (23 Jan. 2022), p. 15539. ISSN: 2071-1050. DOI: 10.3390/su142315539. URL: https://www.mdpi.com/2071-1050/14/23/15539 (visited on 05/24/2025).

[21] Rijad Sarić et al. "Applications of Hyperspectral Imaging in Plant Phenotyping". In: *Trends in Plant Science* 27.3 (Mar. 1, 2022), pp. 301–315. ISSN: 1360-1385. DOI: 10.1016/j.tplants.2021.12.003. URL: https://www.sciencedirect.com/science/article/pii/S1360138521003460 (visited on 05/22/2025).

[22] Michèle R. Slaton, E. Raymond Hunt Jr., and William K. Smith. "Estimating Near-Infrared Leaf Reflectance from Leaf Structural Characteristics". In: *American Journal of Botany* 88.2 (2001), pp. 278–284. ISSN: 1537-2197. DOI: 10.2307/2657019. URL: https://onlinelibrary.wiley.com/doi/abs/10.2307/2657019 (visited on 05/22/2025).

[23] Tingting Tan et al. "Far-Red Light: A Regulator of Plant Morphology and Photosynthetic Capacity". In: *The Crop Journal* 10.2 (Apr. 2022), pp. 300–309. ISSN: 2214-5141. DOI: 10.1016/j.cj.2021.06.007. URL: https://linkinghub.elsevier.com/retrieve/pii/S2214514121001501 (visited on 05/22/2025).

[24] Jose C. Tovar et al. "Raspberry Pi–Powered Imaging for Plant Phenotyping". In: *Applications in Plant Sciences* 6.3 (2018), e1031. ISSN: 2168-0450. DOI: 10.1002/aps3.1031. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/aps3.1031 (visited on 05/24/2025).

[25] Vincent Vadez et al. "LeasyScan: A Novel Concept Combining 3D Imaging and Lysimetry for High-Throughput Phenotyping of Traits Controlling Plant Water Budget". In: *Journal of Experimental Botany* 66.18 (Sept. 2015), pp. 5581–5593. ISSN: 0022-0957, 1460-2431. DOI: 10.1093/jxb/erv251. URL: https://academic.oup.com/jxb/article-lookup/doi/10.1093/jxb/erv251 (visited on 05/23/2025).

[26] Cyrille Violle et al. “Let the Concept of Trait Be Functional!” In: *Oikos* 116.5 (2007), pp. 882–892. ISSN: 1600-0706. DOI: 10.1111/j.0030-1299.2007.15559.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.0030-1299.2007.15559.x (visited on 05/20/2025).

[27] Rafael Massahiro Yassue et al. “A Low-cost Greenhouse-based High-throughput Phenotyping Platform for Genetic Studies: A Case Study in Maize under Inoculation with Plant Growth-promoting Bacteria”. In: *The Plant Phenome Journal* 5.1 (Jan. 2022), e20043. ISSN: 2578-2703, 2578-2703. DOI: 10.1002/ppj2.20043. URL: https://acsess.onlinelibrary.wiley.com/doi/10.1002/ppj2.20043 (visited on 05/24/2025).